

Dwij Joshi

Mechatronics Engineering Student at Toronto Metropolitan University

Scarborough, ON · 437-925-2502 · dwijjoshi14@gmail.com · linkedin.com/in/dwij-joshi-14-tron · github.com/dwijjoshi14

Education

Toronto Metropolitan University

Expected Apr 2029

Bachelor of Engineering in Mechatronics Engineering

Ontario Scholar · Embedded Systems, Data Structures and Algorithms, Digital Logic, Circuit Analysis, OOP, Discrete Mathematics

Technical Skills

Languages: C, C++, Python, JavaScript, HTML/CSS, SQL, Java

Embedded: FreeRTOS, STM32, Arduino, PWM, ADC, UART, I2C, SPI, GPIO, PMOS driver circuits, PlatformIO

Tools and Platforms: Git, Linux, VS Code, MATLAB, KiCad, Tinkercad, Make, Next.js, Supabase, Vercel

Testing: Unit testing, black-box and white-box testing, defect tracking, validation and verification

Experience

Webmaster and Full-Stack Developer

Apr 2025 to Present

Mechatronics Course Union (MCU), Toronto Metropolitan University

- **Sole developer** who designed and shipped www.tmutron.ca from scratch using vanilla HTML, CSS, and JavaScript across 6 pages covering events, team, academics, calendar, and contact, with **zero frameworks** or external dependencies.
- Engineered a **canvas particle** and circuit background system, a 3D coverflow carousel, **RGB-split glitch effects** on page load, a custom HUD cursor, scroll progress bar, and per-section VFX, all hand-coded without libraries.
- Reduced the club's operational cost from \$240 per year to \$20 per year by self-hosting on GitHub Pages with a custom domain, a **92% reduction** from the original agency quote.
- Integrated a live Square e-commerce shop, social platform connections, a light and dark mode toggle, **mobile responsive** layout, and page transition animations.

Web Designer

May 2026 to Jul 2026

Can Crops Inc.

- Built and launched a fully responsive 6-page website for an early-stage agri-tech startup, produced over 30 branded digital assets, and cut future asset production time by roughly **40%** through a **reusable component library**.

Administrative Assistant

Jul 2024 to Aug 2025

Rosalie Hall Mental Health Institution

- Rebuilt the Excel-based client tracking system for 50+ users, cutting report generation from around **3 hours to under 45 minutes**. The updated protocol was adopted across **3 program teams** and improved audit readiness.

Intake Worker

May 2026 to Jul 2026

Settlement Assistance and Family Support Services (SAFSS)

- Managed end-to-end intake and data entry for 30+ weekly cases with **zero database integrity errors** while maintaining confidential records for over 150 active client files.

Projects

RTOS Task Scheduler

Apr 2026 to May 2026

C11 FreeRTOS Queue Semaphore Mutex STM32-Portable

- Architected a 3-task **FreeRTOS** system across SensorTask, ControlTask, and DisplayTask with **priority-based scheduling**, typed queue inter-task communication, a binary semaphore for critical temperature alerting, and a mutex for shared resource protection.
- Built a **POSIX simulation layer** that maps the full FreeRTOS API to pthreads so identical code runs on both a host machine and STM32. Porting to hardware requires only 3 HAL function changes with **zero modifications** to any task logic.
- Validated all threshold decision logic in **isolated C unit tests** before touching hardware, fully decoupling algorithm correctness from hardware dependency.

IMU-Based Robotic Arm Controller

May 2026 to Present

C++ STM32F103 FreeRTOS MPU-6050 I2C PWM

- Implementing a **complementary filter** that fuses MPU-6050 accelerometer and gyroscope data over I2C for **drift-free** real-time angle estimation on STM32F103, with a 3-task FreeRTOS architecture and queue-based IPC.

ClearShift — Full-Stack Hackathon App

Jun 2026

Next.js Supabase Claude API Vercel

- Built and **shipped** a full-stack shift handoff tool for clinical care teams at Mind the Product World Product Day. The app went live on Vercel within the competition window after diagnosing and resolving a Supabase SSR prerender failure through **Suspense boundaries**, runtime environment guards, and **Claude API** integration for AI-generated handoff summaries.

Embedded Heat Management System

Sep 2025 to Dec 2025

C++ Arduino PWM ADC TMP36 PMOS Tinkercad

- Designed a **closed-loop** thermal control system using TMP36 ADC sensing and threshold-based PWM fan control through a **PMOS high-side driver**. Caught and resolved an inverted PWM logic error caused by PMOS switching behavior through **iterative Tinkercad simulation** before any hardware was touched.